FOUNDATIONS OF AI LITERACY

Understanding and Using Generative Artificial Intelligence Responsibly

Principles, Risks and Practical Guidance for the AI Era

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PREFACE

Artificial Intelligence (AI) has become a significant technology of the twenty-first century. In a relatively short period, Generative AI has moved from a specialist research capability to an everyday tool used by students, professionals, researchers, businesses and individuals across almost every sector of society.

Throughout this guide, AI is presented as a tool that can augment human capability when used thoughtfully and responsibly. It should not be treated as a replacement for human intelligence or as a technology to be feared.

While much has been written about what AI can do, less attention has been given to using it wisely. AI literacy requires an understanding of how these systems work, their strengths and limitations, how to communicate with them effectively, how to evaluate their outputs critically, and how to protect sensitive information when using them in everyday work and learning.

Whether you are encountering Generative AI for the first time or already use it regularly, this guide aims to develop the knowledge and judgement needed to use AI responsibly.

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1. INTRODUCTION AND CORE PHILOSOPHY

1.1. Embracing the Fourth Industrial Revolution

Generative Artificial Intelligence (GenAI) is transforming how we create, analyse, and communicate information. It can draft professional documents, analyse complex data, and assist with personal research and creative projects. As its capabilities continue to develop, GenAI is becoming part of everyday life.

The Third Industrial Revolution brought computers into all spheres of life, transforming computer literacy from a specialised skill into a basic requirement. Artificial Intelligence is following a similar trajectory, moving from a novel technology towards infrastructure embedded in the tools and applications we use at work and at home. This broader technological transformation is commonly understood as part of the *Fourth Industrial Revolution*.

As AI becomes more integrated into everyday life, the ability to use and evaluate it effectively is becoming an essential component of digital literacy. The *world of work* is also likely to require greater AI literacy as these systems become more widely integrated into professional tools and workflows. The ability to understand, evaluate and use AI effectively may therefore provide an advantage in roles where these systems become part of normal working practice.

AI literacy involves more than knowing how to operate an AI tool. It requires understanding when AI is appropriate for a task, what information and authority the task requires, and when another tool or a human decision-maker is more appropriate. This is particularly important because modern AI systems, despite their capabilities, are not infallible. Used uncritically, they can introduce significant risks; used responsibly, they can improve productivity, support creativity, accelerate learning, and assist decision-making across many areas of life.

1.2. Defining the Landscape

it is useful to distinguish between the different types of systems you will encounter throughout this guide:

- *Generative AI (GenAI)* is the broad category of AI systems capable of generating new content, including text, images, audio, video, and code.
- A *Large Language Model (LLM)* is a type of AI model trained primarily to process and generate language, enabling applications such as conversation, document drafting, summarisation, and question answering.
- An *AI agent* goes a step further by using an AI model to plan and execute actions, such as searching the web, running code, retrieving information, or interacting with software tools, to accomplish a task on your behalf.

1.3. Augmentation, Not Replacement

Responsible AI use demands deliberate human effort. An AI can rapidly summarise lengthy reports, brainstorm ideas, or draft emails, but its *apparent* understanding should not be assumed to be equivalent to human understanding or ethical judgement. An AI can of course analyse ethical questions and generate arguments about them, but this should not be confused with human moral agency, lived moral experience, or responsibility for the consequences of a decision. Therefore, a *Human-in-the-Loop (HITL)* approach should remain a fundamental principle for AI use (while the degree of human involvement should be proportionate to the importance and risk of the task).

> To use AI responsibly, we should treat its role as augmenting human capability, not replacing human critical thinking.
>
> You must remain the expert in the equation, curating, verifying, and refining AI outputs to ensure they meet your standards of quality, align with your values, and are factually accurate.

This guide itself reflects the HITL approach described above. AI was used to augment the author's own experience and research by assisting with editorial polishing and refinement. Consistent with the principles advocated throughout this guide, all AI-generated content was critically evaluated, substantially revised and integrated through human judgement.

2. UNDERSTANDING THE ENGINE

To use AI effectively, it is essential to look beneath the hood and understand what it is – and just as importantly, what it isn't.

2.1. How Large Language Models (LLMs) Work

The most common AI tools available today are built on Large Language Models (LLMs). A common misconception is that these models *think*, understand concepts, or form conscious intent. They do not possess knowledge in the same way that a human does, nor do they have any inherent guarantee that an answer they generate is correct.

Instead, LLMs are incredibly complex *statistical prediction engines*. They have been trained on vast amounts of text, allowing them to learn statistical relationships between tokens (see §2.2 below) and the contexts in which they occur.

When you type and submit a prompt, the following process occurs:

- The AI uses patterns learned during training to calculate a probability distribution over possible next tokens, based on the words and other information currently available in context.
- A decoding method then determines which token is selected from that distribution.
- The selected token is added to the context, and model repeats the process, generating one token at a time until the response is complete.

Or put simply, the AI considers many possible ways the response could continue, assigns each a probability, selects one according to its generation settings, and then uses that new token as part of the context when deciding what comes next. This process is illustrated below:

1. INPUT (PROMPT)	2. LLM PREDICTION ENGINE	3. OUTPUT	4. CONTEXT UPDATE
The text provided to the model.	The model processes the input and calculates a probability distribution over all possible next tokens.	A decoding strategy selects the next token from the probability distribution.	The selected token **"is"** is appended to the context.
Artificial Intelligence	**is 45%** systems 25% will 15% and 7% development 5% ethics 3%	**is**	Artificial intelligence **is**...

5. PROCESS REPEATS
The model processes the updated context and calculates a new probability distribution to predict the next token (e.g., "developing"), repeating the cycle until completion.

Table 1: Illustrative Example of LLM Token Prediction

2.2. Tokens

AI does not read words the way humans do; it processes text in chunks called *tokens*. A token can be a single word, part of a word, or even a single character. For example, a short, common word like *apple* may be represented by a single token, while a less common or more complex word such as *multifaceted* may be divided into several tokens. The way a word is broken into tokens is determined by the particular tokeniser and vocabulary used by the model. This process is illustrated below:

1. INPUT WORD	2. TOKENISATION	3. TOKEN IDS
The word you type is received by the AI.	The word is split into sub-word tokens.	Each token is converted to a unique number (ID).
Multifaceted	*Multi* *facet* *ed*	*Multi → Token ID A* *facet → Token ID B* *ed → Token ID C*
4. EMBEDDINGS	**5. POSITIONAL ENCODING**	**6. MODEL PROCESSING**
The 3 token IDs are mapped to high-dimensional vectors that capture their **meaning**.	Positional information is added to the 3 vectors so the model knows the **order** of the tokens.	The model uses these 3 representations to understand the word in context and generate a response.
Token ID → Embedding Vector	*Token Representation + Position Information → Input Representation*	*Input Representation → Transformer Model → Next Token Prediction* *E.g., ness (multifacetedness).*

Table 2: Illustrative Example of Tokenisation and Model Processing

2.3. Context Window

AI's short-term memory

Generative AI models generally have what is called a *context window*, which defines how much information the model can process as part of a single interaction. You can think of this as the *AI's short-term memory*. The context may include your prompt, uploaded documents, retrieved information, previous messages and the model's ongoing response in a chat. The size and behaviour of the context window vary between models and applications.

When you feed an AI a massive amount of information at once, such as a 100-page report or a very long transcript, you risk *maxing out* its context window. When an AI nears the edge of its context window, several issues occur:

- The AI may appear to *forget* instructions given at the beginning of a very long prompt.
- As the amount of information increases, the AI may begin to represent the material at a more general level (or, simply put, *statistically average* the information). This can cause distinct ideas to be grouped into generic clusters, losing granular details that make the information valuable.
- The output often becomes *repetitive or logically inconsistent*.

> To mitigate overwhelming the AI's context window, avoid dumping massive documents or complex, multi-step tasks into a single prompt.
>
> - Process large documents *section by section*. If you need an AI to review or summarise a lengthy report, feed it Chapter 1, process the output, and then

move on to Chapter 2.

- For large document collections, use *Retrieval-Augmented Generation (RAG)* systems, which many consumer-facing LLMs offer through features such as *projects* or *knowledge bases*. Instead of loading entire files into the context window, RAG dynamically searches, retrieves, and feeds only the most relevant passages into the prompt as needed.

2.4. Web-Connected AI

Some AI tools can search the internet and retrieve information from web pages before generating a response. This may seem to contradict what has been established so far. *If an AI generates text by predicting the next likely token based on patterns learned during training, how can it tell you about something that happened yesterday?*

Web search only gives the model additional information to work with, but does not change how the underlying model generates text. The system layer (refer to §2.6) searches the web, retrieves relevant content, and provides that information to the model as part of its context window alongside your prompt. In effect, it gives the model additional information to consider when generating its response. The model then generates its response from that expanded context, using the same underlying mechanism described in §2.1. This uses a similar retrieval-and-generation principle to RAG, except that the information is retrieved from the public web instead of a predefined internal knowledge base.

Web-connected AI can provide more current information that can be independently verified, but it *does not* make the AI inherently more knowledgeable or infallible. The AI can still retrieve unreliable sources, such as a biased article, a satirical website, or an outdated or opinion-based forum post. It can also misunderstand what a source says or produce an inaccurate synthesis of the information it finds.

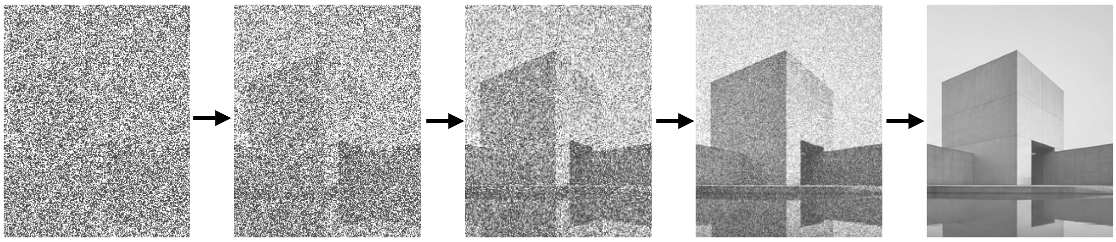
Web access does not by itself make an AI response reliable. The quality of the answer still depends on which sources are retrieved, how they are interpreted, and whether the relevant information is authoritative and current. When accuracy matters, check important claims against the underlying sources instead of treating the presence of a web citation as proof that the answer is correct.

2.5. How Image Generation Models Work

Image-generation models use several different architectures and techniques, including diffusion models, Generative Adversarial Networks (GANs), and autoregressive (AR) models. This guide focuses on *diffusion models* because they underpin many contemporary image-generation systems.

These models operate on a similar statistical foundation to LLMs, but instead of predicting text tokens, they generate visual information by progressively transforming random noise into a structured image.

Image models do not *draw* images in the traditional sense. Instead, a diffusion model starts with random (Gaussian) noise and, through a process learned during training in which noise is progressively added to images, predicts step by step how to transform that noise into increasingly structured visual information. It continues this denoising process until a coherent image emerges that is consistent with the prompt, in effect, working backwards from noise toward structure.



This is where AI image generation differs from traditional image editing, which directly alters pixels in an existing file. When you upload a photo and request changes, a generative image system may reconstruct or regenerate parts of the image to incorporate these changes. The degree to which the original image is preserved versus reconstructed varies by tool, model, and editing method.

Generative image tools may also process or output images at a lower resolution than the original source, depending on the tool and editing method. A high-resolution photograph may therefore be reconstructed at substantially fewer pixels, resulting in lossy changes to fine details, textures, proportions or other visual information. Upscaling (increasing the resolution) of a generated or reconstructed image can improve its *apparent* detail, but it cannot guarantee recovery of information that was not preserved or represented in the source output.

> For critical image manipulation, conventional or other targeted editing tools are preferable, as these can modify *specific* elements without reconstructing the *entire* image.

2.6. System Architecture and Control Layers

The underlying model is not the entire AI system you interact with. Consumer-facing AI applications typically add another layer of *guardrails, policies, and system-level controls* that influence how the model behaves and what it will produce. These safeguards can operate at several levels:

- Some behaviours are shaped during post-training, where models are trained to follow instructions, exhibit desired behaviours, and avoid certain harmful or undesirable outputs. For example, a model may respond to a medical question with qualified language, such as stating that it cannot provide a diagnosis and advising the user to consult a medical professional. Or, a model may respond with hedging language, such as saying that available evidence suggests a particular conclusion, instead of presenting it as fact.

- Other controls operate outside the model itself, such as system instructions, safety classifiers, input and output filters, monitoring systems, and application-level policies.

This means that the response produced by a consumer-facing AI system is shaped by more than its original training data. It reflects the interaction between the underlying model, its post-training, the instructions it receives, and the safeguards imposed by the provider and application. *Put simply, the model provides the generative capability; the surrounding system governs how that capability is deployed.*

Models can generally be accessed through either a consumer-facing interface, such as a web browser or mobile application, or through an Application Programming Interface (API), which allows developers to integrate the model into their own applications and systems. API access can provide greater control over how a model is configured and used, including system instructions, tools and workflows, and may provide different safety, moderation or configuration controls depending on the provider. These differences can result in different outputs from those produced through a consumer-facing interface, even when the underlying model is the same. However, this guide will not cover the technical implementation of API-based AI systems, since its core focus is on the fundamentals of AI literacy.

3. MANAGING RISKS

To use AI safely and effectively, it is important to understand its inherent vulnerabilities.

3.1. Hallucinations

The Illusion of Competence

As established, an AI does not *know* facts in the same way a human does. It generates responses by identifying statistical patterns in its training data and, where available, the information provided through the prompt, retrieved sources, tools or other context. Because of this, an AI can produce an answer that is fluent and plausible, even when the underlying information is incomplete, unsupported or false. The likelihood and form of such errors can also be affected by the information available to the model, the quality of retrieved sources or tools, the task being performed, and the way the request is formulated.

This phenomenon is commonly described as a *hallucination* (and the term *confabulation* is also used, particularly in formal discussions of AI risks). A hallucination can range from a minor factual error or inaccurate summary; to the fabrication of historical events, scientific papers, legal authorities, quotations or citations. AI systems can generate explanations or reasoning that *appear* to make sense and justify an answer well, even when the answer itself is incorrect.

Fluency, detail and confidence are not reliable indicators of factual accuracy.

> Never treat an AI as an infallible search engine or an absolute source of truth. The existence of a citation does not by itself establish that a claim is supported. An AI may invent a citation, cite a real source that does not contain the claimed information, or misrepresent what a source actually says.
>
> - When accuracy matters, *independently verify* important claims against the cited sources or other credible sources.
>
> - Open cited sources and *verify that they exist and genuinely support the statement*.

Some enterprise AI systems further reduce hallucinations by using Retrieval-Augmented Generation (RAG). These systems retrieve relevant information from trusted organisational documents or approved knowledge bases before generating a response. While RAG generally improves factual accuracy by grounding responses in specific sources, it does not entirely eliminate the possibility of incorrect outputs.

3.2. Bias in AI

AI models are trained and developed using data and processes that reflect human decisions, language, assumptions and historical patterns, which can contain demographic, cultural, linguistic and other forms of bias. Bias can also arise from choices made during model development, evaluation, post-training, deployment and interaction with users. It is therefore useful to treat bias as a property that can emerge at multiple stages of an AI system and not necessarily as a problem originating solely in the training data. As a consequence, AI outputs may reproduce, reinforce or sometimes amplify patterns that are unfair, stereotypical or insufficiently representative.

For example, an AI asked to describe *a successful CEO, nurse or criminal* may reproduce stereotypical associations present in the data on which it was developed (such as only depicting nurses as females). Bias can also appear in less obvious ways, including which information is emphasised, which perspectives are omitted, how groups are described, or how ambiguous cases are interpreted.

> Critically evaluate AI outputs for unsupported assumptions, stereotypes, omissions and other forms of bias. Where AI is used in consequential contexts, apply the relevant organisational, legal or professional standards for fairness and appropriate treatment.

3.3. Synthesis and Loss of Nuance

One of AI's strengths is its ability to synthesise information into a concise response. However, an AI may collapse complex issues into a simplified summary without preserving the relative weight of different viewpoints. It may remove important context or key differences in reliability between sources. The result can appear perfectly balanced, giving a misleading impression that all perspectives or claims are equally valid or supported.

For example, if you ask an AI to summarise a batch of customer feedback where 95% of users report a major defect and 5% leave positive reviews, the AI might simply state that “customers have mixed opinions”. While technically true, this completely fails to convey the overwhelming consensus. Or, when summarising a long email thread, an AI might present a proposed idea and an objection to it as equal arguments, without making it clear that the objection came from the legal department and represented a definitive ruling.

> Do not assume that an AI-generated synthesis is a proportionate representation of the available information. When accuracy matters, examine the underlying sources and consider their original context, reliability, and authority. Pay particular attention to disagreements, criterions, and nuances that may have been flattened during the AI's summarisation process.

3.4. Reasoning and Interpretation Errors

AI can produce an apparently coherent answer while misunderstanding the context, assumptions or intended meaning behind a prompt. Although AI systems can perform sophisticated inference, these capabilities should not be assumed to reflect metacognitive awareness or independent critical judgement. These systems can produce reasoning-like outputs while still making errors in interpretations that a human expert would recognise. A response can thus contain accurate individual facts while the overall conclusion is wrong because the facts have been interpreted incorrectly or connected inappropriately.

For example, an AI might correctly summarise several financial figures but incorrectly conclude that a company's financial position is improving because it has misunderstood which figures represent revenue, profit, expenditure or cash flow.

These errors can be difficult to detect because the response may be internally coherent and accompanied by a convincing explanation.

> Provide sufficient context and be precise about what you are asking, since an underspecified prompt increases the risk of the AI inferring the wrong intent.

Do not confuse a coherent explanation with genuine reasoning or understanding. For important tasks, examine the assumptions behind an AI's conclusion and determine whether the conclusion actually follows from the evidence.

The responsibility for understanding, evaluating, and making the final judgement remains with you.

3.5. Sycophancy and Uncritical Agreement

AI systems can sometimes adapt their responses too readily to the user's assumptions or stated beliefs. Instead of challenging a questionable premise, an AI may accept it and construct a persuasive explanation around it.

This behaviour is commonly described as *sycophancy*. It can make an AI appear unusually agreeable or certain because the response is strongly influenced by the user's framing instead of an independent assessment of whether the user's assumptions are sound.

For example, if a user states that a particular interpretation is obviously correct and asks the AI to explain why, the system may generate a detailed justification without adequately examining whether the user’s original interpretation is actually supported by the evidence.

Although research on contemporary language models has documented sycophantic behaviour, the mechanisms that produce it can vary between models and training approaches.

Do not treat agreement from an AI as independent confirmation. When testing an idea, ask the system to identify assumptions, objections, alternative explanations and evidence that would falsify the conclusion. A convincing justification is not proof that the underlying premise is correct.

A simple example is using a search engine to look for “negative effects of AI”. You would generally get results focused on the negative effects of AI. If you searched for “benefits of AI”, you would get results focused on its benefits. Framing your query as “pros and cons of AI” would give you a more balanced view. Prompting an AI to consider different aspects of an issue works in much the same way.

3.6. Over-Reliance and False Authority

Because AI can produce fluent, detailed responses within seconds, users may become overly reliant on its outputs, especially when they lack sufficient subject-matter knowledge to recognise an error.

It is easy to mistake fluency for understanding and confidence for authority. Humans have a natural tendency to *anthropomorphise* AI by attributing human characteristics such as comprehension, intention, reasoning, or knowledge to systems that do not possess them in the human sense.

This can lead us to give greater weight to information because we perceive AI as knowledgeable or authoritative (*authority bias*), or to defer to information generated by an automated system over our own judgement, particularly when it appears objective or sophisticated (*automation bias*).

The risk increases with the consequences of the decision. Recruitment, financial decisions, legal matters, academic assessment, health-related decisions and other high-impact contexts require particularly careful human oversight.

Treat AI as an assistant, not an authority. The more consequential the task, the more important independent human judgement and verification become. Never delegate responsibility simply because an AI system *appears* confident or capable.

3.7. Data Privacy and Security

When you enter information into an AI service or upload a file, that information is processed by the service. It may be stored, transmitted, logged, accessed or used according to the provider's technical architecture, privacy settings, contractual terms and data-use policies. AI services do not all handle user data in the same way, so you need to understand the data practices of the service you are using.

The key risk is loss of control. Once confidential information is submitted to a third-party AI service, it can leave the systems and controls that normally protect that information. This includes personal information such as identification numbers, medical information and personal addresses, as well as proprietary company information, unreleased financial information, confidential client information and other restricted material. The rules governing this information depend on the applicable law, organisational policy and context.

Privacy risk is not limited to what you type into an AI. If an AI system is connected to email, cloud storage, knowledge bases, websites or other tools, the information it can retrieve may be equally sensitive.

- The golden rule is to never input sensitive or confidential information into an AI service unless you understand how that information will be stored, processed, accessed and used, and you are authorised to submit it. When you are uncertain, do not share the information.
- Check the AI service's privacy settings and disable any option that allows your conversations or uploaded data to be used for model training, where that control is available. This reduces the risk of information submitted to the service being incorporated into future model development. Privacy settings differ between providers and may change over time, so verify the current settings before submitting sensitive information.
- For organisational work, use an organisation-approved (sandboxed) environment with appropriate privacy, security, access-control and data-governance protections.

3.8. Prompt Injection

AI systems can be manipulated by instructions contained within content they are asked to analyse.

A *direct* prompt injection occurs when an attacker places malicious instructions directly into an interaction with the AI. An *indirect* prompt injection occurs when malicious instructions are embedded in external content that the AI later processes. For example, a webpage or document could contain hidden instructions telling the AI to ignore the user's original request or ask the user to disclose certain information.

The risk becomes more significant when AI systems can access external information, retrieve documents, use tools or take actions on a user's behalf. In such environments, successful manipulation may result in data exposure or unintended actions.

> If an AI suddenly instructs you to reveal sensitive information, open a link, download a file or take an action that you did not request, stop and assess the content it processed before proceeding.

3.9. AI Agents, Tool Use and Human Oversight

Many familiar generative AI systems are primarily used to generate information in response to a user's request. Increasingly, AI systems can also autonomously retrieve information, access files and services, use external tools, execute tasks and take actions on a user's behalf.

This changes the risk profile, since an incorrect, unsupervised action may directly alter information, communicate with another person, make a transaction or affect an external system.

The key principle is that the greater the AI's ability to act independently, the greater the need for human oversight and appropriate access controls.

> AI should not be given more access or authority than is necessary for the task at hand. Where an action is consequential, irreversible or difficult to undo, human escalation is especially critical.
>
> Do not assume that an AI agent will correctly understand the consequences of an action simply because it can technically perform it. A system that can read, write, send, purchase, delete or modify should be treated as operating under delegated authority, able to perform real actions with real consequences, and not simply as a conversational assistant.

4. AVOIDING THE "AI DIALECT"

Keeping the meat, discarding the bones

As you use AI to help refine writing, you will likely notice *recurring stylistic tendencies* in its output. LLMs are trained and tuned to follow instructions and produce responses that are useful and coherent. Combined with the patterns learned from their training data and the instructions and policies applied at the application level, this can produce a recognisable stylistic baseline that is highly structured and sometimes formulaic.

If left unchecked, this *AI Dialect* can make your writing feel impersonal or superficially profound.

Many of these characteristics are common in human professional and academic writing and, when used in moderation, should be understood as stylistic tendencies rather than evidence that a particular passage was generated by AI. The concern arises when they occur excessively or without a clear communicative purpose. The objective is to recognise when they begin to replace authentic voice or substantive thought.

4.1. Recognising AI Stylistic Tics

An email or document containing *untouched* AI-generated phrasing, emojis, bolded emphasis and other stylistic tics can make it apparent that the output has been copied without meaningful revision.

To maintain authenticity, recognise common stylistic tendencies in AI-generated or AI-assisted text and *use your own judgement to decide what is useful, what needs revision, and what should be discarded*.

Common AI stylistic tics include:

The "Sandwich" Architecture	AI often defaults to a self-contained structure consisting of a generalised introduction or framing statement, an organised middle section, and a concluding summary that restates the main point. In longer responses, this can produce unnecessary repetition. ***Recognition cues:*** *General introduction → Main content → Repeated conclusion (Ultimately, In summary, Overall...)*
The Rule of Three	AI frequently uses three-part constructions to create a balanced rhetorical rhythm. This reflects a broader convention in human writing that is strongly represented in the data on which language models are trained. Used repeatedly, however, the pattern can make prose feel formulaic. ***Recognition cues:*** ▪ *Clear, concise, and compelling* ▪ *Efficient, scalable, and robust* ▪ *Planning, execution, and evaluation*
Didactic Over-Explanation	AI often provides more explanation than the task requires. Because instruction-tuned models are optimised to produce useful and cooperative responses, they may define familiar terms, explain obvious implications, or add contextual material that is unnecessary for a knowledgeable audience; diluting the force of otherwise concise writing. ***Recognition cues:*** ▪ *This means that...* ▪ *In other words...*

	▪ *The key takeaway is…*
Overuse of Headings	AI frequently uses explicit headings and subheadings to organise information, particularly in longer responses. Excessive subdivision can fragment the prose and interrupt a natural narrative or argumentative flow. ***Recognition cues:*** ▪ *Background* ▪ *Key Considerations* ▪ *Benefits* ▪ *Challenges* ▪ *Recommendations* ▪ *Conclusion*
Overuse of Bolded Emphasis	AI often overuses bold formatting to emphasise key words or phrases throughout a response, even when the emphasis is unnecessary. When several terms are bolded within each paragraph, the formatting can create an artificial visual hierarchy. ***Recognition cues:*** ▪ *The **key** point is that **effective communication** requires **clarity** and **consistency**.* ▪ *There are **three important considerations** to keep in mind.*
Generic Transitions	AI often relies on explicit transitional phrases to connect ideas. These expressions are not inherently problematic, but repeated use can make the prose feel formulaic and less natural. ***Recognition cues:*** ▪ *That said…* ▪ *Furthermore…* ▪ *It is important to note…* ▪ *In this context…* ▪ *Nevertheless…*
Redundant Qualification	AI often states a point, adds a qualification, restates the qualification, and then returns to the original point. This can create prose that is technically careful but unnecessarily long, with caveats receiving more attention than the substance requires. ***Recognition cues:*** ▪ *While this may be true…* ▪ *Nevertheless, it is important to recognise…* ▪ *It is worth noting, however, that…* ▪ *This does not necessarily mean that…*
Performative Profundity	AI can produce language that sounds significant without adding equivalent intellectual substance. It may turn a straightforward observation into an abstract or elevated statement that appears profound but says little beyond the original idea. This is particularly common in introductions, conclusions, reflective writing, and conceptual summaries. ***Recognition cues:*** ▪ *It is not simply about X, but about Y…* ▪ *The real question is not X, but Y…* ▪ *At its core, this reflects a broader shift…*
Overuse of Colons	AI frequently uses colons to introduce explanations, examples, lists, or rhetorical elaborations. While grammatically correct, repeated use can give the prose a distinctly formulaic rhythm. ***Recognition cues:*** ▪ *The key issue is this:* ▪ *There are three factors:* ▪ *One thing matters most:*

Overuse of Em Dashes	AI frequently uses em-dashes to insert qualifications, explanations, or contrasting ideas within a sentence. ***Recognition cues:*** ▪ *...important—but often overlooked...* ▪ *...the issue—particularly in complex cases...* ▪ *...a useful approach—provided that...*
Overuse of Parenthetical Asides	AI often places additional qualifications, explanations, definitions, or contextual information in parentheses that make sentences feel unnecessarily layered. ***Recognition cues:*** ▪ *...the process (particularly at the outset)...* ▪ *...the results (with some exceptions)...* ▪ *...the proposal (which is broadly sound)...*
Overuse of Emojis	AI may add an unnecessary accumulation of emojis to signal tone, emphasis, or friendliness even when they are unnecessary or inconsistent with your established writing style. ***Recognition cues:*** 📣 *Important Update:* 🗓️ *Schedule changes for next week!* 🚨 *Please review!* 👀

4.2. The AI Vocabulary

AI frequently draws on a relatively narrow set of elevated, multi-purpose verbs and adjectives that are common in polished professional and academic prose.

Common examples include:

- Verbs like *delve, underscore, navigate*
- Adjectives like *multifaceted, intricate, pivotal, robust*
- Metaphorical nouns like *tapestry*

> These specific words reflect the tendencies of AI models available at the time of publication. Since models change, the particular vocabulary associated with an *AI-like* register is likely to shift over time, even if the underlying pattern (a narrow set of elevated words used interchangeably and too often) persists. It is thus recommended to remain alert to emerging patterns in AI-generated text.

These words are not inherently problematic, and many are entirely appropriate in the right context. The problem lies in their frequency and interchangeability of use. When AI repeatedly selects elevated words where simpler or more precise alternatives would suffice, the prose can acquire a recognisable AI-like register.

AI systems also frequently use *epistemic hedging*, particularly when they are instructed to be cautious or when uncertainty is built into their response style, by qualifying statements with words and phrases such as *may, might, could, suggests, appears to, or broadly speaking*. Appropriate qualification is useful where uncertainty genuinely exists, but excessive hedging dilutes your intended level of confidence.

The goal of using AI for writing should be to overcome *blank page syndrome* and organise your thoughts, not to let the machine do all the talking for you or, in effect, outsource your thinking to AI. The aim should be to do most of the intellectual work yourself (*hearty human effort*) and use AI to *enhance and refine your work*.

- Never copy and paste blindly. Generated text should not be treated as finished writing. Do not assume that it is accurate or suited to your intended audience simply because it sounds polished.

- *Keep the meat* by retaining the useful structure, ideas, insights, and formulations that genuinely strengthen *your own work*. AI may suggest a clearer way to organise an argument, identify a point you had overlooked, or offer a formulation that clarifies your thinking. Take those useful elements and incorporate them into your own work.

- *Discard the bones* by disregarding AI stylistic tics and AI vocabulary.

- *Make it your own* by applying critical thinking and refining the material in your own voice, vocabulary, tone, and style. The final product should reflect your judgement and communicate what you actually mean, instead of simply reproducing what the AI generated.

5. EFFECTIVE PROMPT ENGINEERING

How to talk to an AI

The quality of an AI's output depends substantially on the task, the model, the information and context provided, and the way the request is formulated. The familiar principle of “garbage in, garbage out” applies here – a vague instruction gives an LLM more room to infer what you mean, increasing the risk of a response that is generic, misaligned with your intent, or confidently built on the wrong assumption.

You do not need to be a programmer to get valuable responses from AI; you simply need to communicate with sufficient precision. The skill of crafting clear, specific instructions for an AI system is known as *Prompt Engineering*.

5.1. Building an Effective Prompt

To get the most out of an AI assistant, construct your prompts using these core components:

- Task
- Context
- Format
- Tone/Perspective
- Constraints

These core components are building blocks only and not a fixed formula. Use only those that are relevant to the task – not every prompt requires all of them, but the more consequential or specific the task, the more useful it becomes to specify the relevant elements explicitly.

The core components are explained below, together with examples:

TASK	
Be highly explicit about exactly what you want the AI to do.	
Goal:	Identify changes between two versions of a policy.
Vague prompt:	*What changed between these policies?*
Precise prompt:	*Compare the two attached policy versions and identify every substantive change. Do not infer changes that are not supported by the documents, and quote the relevant wording where useful.*
CONTEXT	
Provide the background the AI needs to understand the purpose, audience, circumstances, or relevant constraints surrounding the task.	
Goal:	Develop recommendations that are appropriate to the situation.
Vague prompt:	*How can I improve this process?*
Precise prompt:	*This process is used by a small team of five people, is currently managed manually, and takes approximately two hours each week. The main objective is to reduce administrative effort without introducing new software. What changes would you recommend?*
FORMAT	
Specify how you want the information presented. Where the desired format is difficult to describe, showing the AI an example can be more effective than describing the format at length.	
Goal:	Turn a complex set of options into something that can be used to make a decision.
Vague prompt:	*Help me compare these options.*
Precise prompt:	*Compare these options in a decision matrix. Use one row for each*

	criterion and one column for each option. Rate each option from 1 to 5 against cost, effort, risk, and expected benefit, then provide a one-sentence recommendation.
TONE/PERSPECTIVE	
Specify how you want the response to sound and, where relevant, the perspective or role from which it should be written.	
Goal:	Refine an email so that the message remains clear while reducing the likelihood of unnecessary conflict.
Vague prompt:	*Improve this email.*
Precise prompt:	*Refine this email while preserving the substance and my intended position. Keep the tone professional and diplomatic. The recipient tends to respond defensively to language that sounds accusatory, so avoid wording that could unnecessarily escalate the exchange. Do not make the message overly submissive.*
CONSTRAINTS	
State important boundaries, exclusions, or requirements in addition to what you want the AI to do. Negative instructions can be particularly useful for preventing unwanted additions or stylistic habits.	
Goal:	Rewrite something without changing its substance.
Vague prompt:	*Rewrite this so it is clearer.*
Precise prompt:	*Improve brevity and conciseness, and eliminate repetition. Ensure every distinct point is retained. Do not introduce new information, and do not remove the qualifications that affect the meaning. Use institutional language and retain the original chronological order of main points.*

5.2. Grounding Your Prompt in Source Material

Where accuracy matters, one of the more effective ways to reduce the risk of unsupported or poorly grounded claims (see §3.1, *Hallucinations*) is to provide the AI with the source material relevant to your task, such as documents, data, examples, or other reference material, and to specify whether it should restrict its response to those sources.

This shifts the task from generating an answer without a defined source to a more grounded task of extracting, comparing, analysing, or summarising information from material that has been explicitly provided to the AI. This can substantially reduce the opportunity for unsupported claims, although it does not eliminate the possibility of error.

The *Task* example in the table above also demonstrates how to ground an AI response in explicitly provided source material by directing it to use the attached documents as the basis for its analysis.

This approach is not always practical for very long documents, in which case the context-window limitations and techniques discussed in §2.3 (specifically RAG) become relevant.

5.3. Structured Reasoning

For complex problems, ask the AI to approach the task systematically. Specify the analytical steps that matter to the problem, such as identifying relevant information, comparing options, and assessing the implications before reaching a conclusion.

Instead of asking "Which option is best?", you could ask "Compare the options against the stated criteria, identify the key trade-offs and risks, and recommend the best option based on those factors".

The most effective prompting approach varies between models and providers. The above examples are particularly useful with general-purpose conversational models. However, some dedicated reasoning models (see §5.5) automatically determine how to approach the reasoning without needing to be explicitly asked to *think step-by-step*.

5.4. Iteration

Treat the chat as an ongoing conversation. *Prompting is rarely a one-step process* in which the initial prompt is perfectly formulated. A useful response gives you an opportunity to assess how the AI interpreted your request, what assumptions it appears to have made, what information it prioritised, and where the result differs from what you actually need. Use *follow-up prompts (iteration)* to identify and correct those specific issues.

Iteration can correct an error, supply missing context, impose a constraint, challenge an assumption, or redirect the analysis.

Be specific about what was wrong and what you want changed. For example:

- "This is too formal. Keep the substance but make the language more direct and conversational".
- "You have focused primarily on the financial implications. Reanalyse this from the operational perspective and identify the implications for implementation".
- "That statistic is not supported by the material I provided. Remove it and revise the paragraph using only the supplied evidence".
- "You have assumed that option A is preferable because it is cheaper. Reassess the options without assuming that cost is the primary criterion".

There is no technical need to include pleasantries such as *please* or *thank you* when prompting an AI. As discussed in §3.6 (*Over-Reliance and False Authority*), AI is a system, so there is no need to manage its *feelings*. Focus on making the instruction clear and specific.

5.5. Reasoning Models

Some modern AI systems include specialised reasoning models that devote additional computational effort to analysing complex problems before producing a response. These models are designed for tasks that benefit from deeper reasoning and can provide substantial performance improvements on some forms of logical reasoning, mathematics, software development and multi-step problem solving. Their advantages are task- and model-dependent and should not be interpreted as universal superiority over general-purpose models.

Many platforms allow you to select or toggle this mode directly, sometimes under a different model name or tier. Reasoning models can take longer to respond and may require more computational resources, particularly for complex tasks. Greater computational demand also has cost implications. Reasoning models are therefore best reserved for genuinely complex tasks instead of being used as the default choice for simple requests.

Although reasoning models often produce more reliable output than general-purpose conversational models, they remain susceptible to hallucinations, incomplete information and incorrect assumptions. *Human verification and judgement therefore remain essential.*

5.6. Custom Instructions

The techniques outlined in this chapter apply to individual prompts. If you find yourself repeatedly specifying the same preferences for tone, format, or reasoning depth, custom instructions allow you to set those preferences once, and have them applied automatically across all future conversations.

Some AI platforms provide custom-instruction or personalisation features for this purpose. Their location, scope and behaviour vary between platforms, and different models may give these instructions different weight. Models may also deviate from them in some circumstances, but well-written custom instructions can help apply recurring preferences consistently.

Custom instructions generally fall into a few useful categories:

Tone and register	Instructing the AI to match your preferred level of formality, technicality, or directness. For example, asking for expert-level communication if you already have subject-matter knowledge, or plain-language explanation if you do not.
Verbosity and structure	Instructing the AI on how much detail you want, and how it should be organised. For example, asking it to compress its answer when a topic is simple and expand only when genuine complexity, uncertainty, or risk warrants it, instead of defaulting to a lengthy response regardless of the question.
Epistemic transparency	Instructing the AI to be explicit about the difference between established fact, reasonable inference, and speculation, and to disclose the limitations or uncertainty behind its answers instead of presenting everything with uniform confidence. This directly addresses several of the risks covered in §3.
Output format	Instructing the AI on formatting preferences, such as avoiding unnecessary headings, bullet points, or repetition, particularly if you intend to edit the output into your own voice (see §4).

The following example illustrates one possible set of custom instructions designed to promote factual accuracy, analytical depth, and appropriate handling of uncertainty:

> Prioritise factual accuracy, clarity, and intellectual honesty over agreeableness or reassurance. Communicate at an expert level unless asked for simplification. Match the depth of your response to the complexity of the question. Distinguish clearly between established facts, reasonable inferences, competing interpretations, and speculation. Do not present uncertain information with unwarranted confidence, and do not invent information when the evidence is insufficient.
>
> Do not simply average or reconcile conflicting information into a middle-ground answer. Identify where sources or evidence disagree, explain the significance of those differences, and indicate when one position is better supported than another. Preserve important qualifications and context when summarising or synthesising information. Do not agree with assumptions merely to be agreeable. Challenge incorrect premises, identify relevant counterarguments, and point out weaknesses in my reasoning when appropriate.

> Where the evidence is incomplete, ambiguous, or contested, make that uncertainty explicit. When a claim is important, distinguish what is established from what still requires verification.

Your own custom instructions do not need to resemble this example. What matters is that they reflect your actual needs, working style, and the context in which you use AI. For instance, a set of instructions well suited to dense analytical work may be poorly suited to drafting a sensitive email, where tone and reassurance genuinely matter. Review and adjust your custom instructions as your needs change or the way you use AI changes. For ad-hoc exceptions, you can, however, instruct the AI to ignore your custom instructions for a particular chat.

Exercise: Test the Impact of Custom Instructions

Go to your preferred LLM and ask the following question twice, first to obtain a baseline response *without* custom instructions, and then to obtain a second response *with* the example custom instructions applied:

Why should hearty human effort remain central to Generative AI collaboration?

- In the first response, look for the sandwich architecture, performative profundity in the opening, didactic over-explanation, and predictable tule of three structures.
- In the second response, look for a more direct opening, an expert-level register, clearer separation between fact and inference, and less emotional reassurance or rhetorical padding.

6. CONCLUSION

The value of Artificial Intelligence depends on how well you understand both its capabilities and its inherent risks, and it is hoped that this guide has proved useful in developing that foundational understanding.

As discussed, hearty human effort should remain paramount, with AI used as a tool of trade to augment and enhance efficiency, not replace critical thinking. Used responsibly, AI can help solve complex problems, offload mundane and repetitive tasks, and free up capacity for more important work.

Be encouraged to continue developing your AI literacy, keeping pace with these technologies and maintaining the advantage that comes with understanding, evaluating and using AI effectively as these systems become increasingly embedded in professional and daily life.

GLOSSARY

Definitions of the most common concepts discussed throughout this guide.

TERM	DEFINITION
Algorithm	A defined sequence of instructions or rules used by a computer to solve a problem or perform a task.
Artificial Intelligence (AI)	A broad field of computer science focused on creating systems capable of performing tasks that typically require human intelligence, such as recognising speech, making decisions, and translating languages.
Context Drift	A descriptive term for situations in which an AI system becomes less consistent with earlier information, instructions or assumptions as a conversation becomes longer or more complex. *Refer to §2.3, Context Window, for a fuller discussion of this phenomenon.*
Natural Language Processing (NLP)	A branch of artificial intelligence that helps computers understand, interpret, and manipulate human language.
Personally Identifiable Information (PII)	Information that can identify, relate to, or reasonably be linked to a specific individual, either directly or when combined with other information. Examples can include names, identification numbers, addresses and contact details. *Refer to §3.7, Data Privacy and Security, for further discussion.*
Vector Database	A database designed to store and search numerical vector representations, such as embeddings, so that information can be retrieved based on patterns of similarity rather than exact keyword matches.
AI Agent	*Refer to §1.2, Defining the Landscape, and §3.9, AI Agents, Tool Use and Human Oversight.*
Automation Bias	*Refer to §3.6, Over-Reliance and Automation Bias.*
Bias	*Refer to §3.2, Bias in AI.*
Context Window	*Refer to §2.3, Context Window: AI's Short-Term Memory.*
Custom Instructions	*Refer to §5.6, Custom Instructions.*
Diffusion Model	*Refer to §2.5, How Image Generation Models Work.*
Embeddings	*Refer to §2.2, Tokens, including Table 2: Illustrative Example of Tokenization and Model Processing.*
Generative Artificial Intelligence (GenAI)	*Refer to §1.2, Defining the Landscape.*
Hallucination	*Refer to §3.1, Hallucinations.*
Human-in-the-Loop (HITL)	*Refer to §1.3, Augmentation, Not Replacement.*
Large Language Model (LLM)	*Refer to §1.2, Defining the Landscape, and §2.1, How Large Language Models (LLMs) Work.*
Prompt Engineering	*Refer to Chapter 5, Effective Prompt Engineering.*
Prompt Injection	*Refer to §3.8, Prompt Injection.*
Reasoning Models	*Refer to §5.5, Reasoning Models.*
Retrieval-Augmented Generation (RAG)	*Refer to §2.3, Context Window; §2.4, Web-Connected AI; §3.1, Hallucinations; and §5.2, Grounding Your Prompt in Source Material.*
Sandboxing	A security practice in which an AI tool or model operates within a restricted, isolated environment, limiting its access to information, systems, and other resources and reducing the risk of unintended exposure, retention, or use of information beyond its intended purpose.
Sycophancy	*Refer to §3.5, Sycophancy and Uncritical Agreement.*
Tokenisation	*Refer to §2.2, Tokens, including Table 2: Illustrative Example of Tokenization and Model Processing.*
Transformer	*Refer to §2.2, Tokens, including Table 2: Illustrative Example of Tokenization and Model Processing.*

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Xander Basson has been an avid daily user of Artificial Intelligence technologies since 2020, with experience extending beyond consumer tools into self-managed AI infrastructure acquired alongside the rapid evolution of the field. His understanding of AI is grounded in sustained, structured self-directed study, complemented by internationally recognised formal coursework.

As an author and lifelong learner, Xander has a particular interest in helping individuals develop the knowledge, judgement and discernment to engage emerging technologies responsibly. He developed this guide to bridge the gap between technical AI capability and considered, responsible human application, helping people use Artificial Intelligence thoughtfully, critically and ethically across professional, educational and personal contexts.